Polarization vectors $\hat{e}_{k\sigma}$, $\sigma = 1, 2$, are built

from real pair $(\varepsilon_{k1}, \hat{\varepsilon}_{k2})$

$$\begin{cases} \hat{k} \cdot \hat{\varepsilon}_{k\sigma} = 0 & \text{transversality} \\ \hat{\varepsilon}^*_{k\sigma} \cdot \hat{\varepsilon}_{k\sigma'} = \delta_{\sigma\sigma'} & \text{orthonormality} \\ \hat{\varepsilon}_{k1} \times \hat{\varepsilon}_{k2} = \hat{k} & \text{right-handedness} \end{cases}$$

Thus $(\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2}, \hat{k})$ form right-handed, orthogonal, Cartesian basis.

In general $\hat{e}_{k\sigma}$ are complex, and can be build from two states of orthogonal <u>linear</u> polarization $\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2}$.
Note that $\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2}$ are real, and $\hat{\varepsilon}_{k1} \times \hat{\varepsilon}_{k2} = \hat{k}$.
Then $\begin{pmatrix} \hat{e}_{k1} \\ \hat{e}_{k2} \end{pmatrix} = U \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix}$, where $UU^+ = U^+U = 1$.
↳ unitary matrix

$$\begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix} \cdot \underbrace{(\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2})}_{} = \begin{pmatrix} \hat{\varepsilon}_{k1} \cdot \hat{\varepsilon}_{k1} & \hat{\varepsilon}_{k1} \cdot \hat{\varepsilon}_{k2} \\ \hat{\varepsilon}_{k2} \cdot \hat{\varepsilon}_{k1} & \hat{\varepsilon}_{k2} \cdot \hat{\varepsilon}_{k2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

here $\begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix}^+ = \left(\begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix}^T \right)^* = \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix}^T = (\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2})$

is just a transpose of real vector $\begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix}$.

Then $\begin{pmatrix} \hat{e}_{k1} \\ \hat{e}_{k2} \end{pmatrix} \cdot \begin{pmatrix} \hat{e}_{k1} \\ \hat{e}_{k2} \end{pmatrix}^+ = U \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix} \cdot \left(U \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix} \right)^+ =$

$$= U \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix} \cdot \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix}^+ U^+ = U \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix} \cdot (\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2}) U^+ =$$

$$= U \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} U^+ = UU^+ = \hat{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

General form of unitary matrix U

$$U = \frac{1}{\sqrt{1+|d|^2}} \begin{pmatrix} 1 & d \\ -d^* & 1 \end{pmatrix} \quad \text{where } d = \underline{\text{complex number}}$$

$$U^+ = \frac{1}{\sqrt{1+|d|^2}} \begin{pmatrix} 1 & -d \\ d^* & 1 \end{pmatrix}$$

then
$$\begin{pmatrix} \hat{e}_{k1} \\ \hat{e}_{k2} \end{pmatrix} = \frac{1}{\sqrt{1+|d|^2}} \begin{pmatrix} 1 & d \\ -d^* & 1 \end{pmatrix} \begin{pmatrix} \hat{\varepsilon}_{k1} \\ \hat{\varepsilon}_{k2} \end{pmatrix} = \frac{1}{\sqrt{1+|d|^2}} \begin{pmatrix} \hat{\varepsilon}_{k1} + d\,\hat{\varepsilon}_{k2} \\ -d^*\hat{\varepsilon}_{k1} + \hat{\varepsilon}_{k2} \end{pmatrix} \quad (1)$$

(a) For $d = \text{real}$, $\{\hat{e}_{k1}, \hat{e}_{k2}\}$ that we obtain from (1) are just real vectors of linear polarization rotated with respect to $\{\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2}\}$.

(b) For $d = i$
$$\begin{cases} \hat{e}_{k1} = \frac{1}{\sqrt{2}}\left(\hat{\varepsilon}_{k1} + i\,\hat{\varepsilon}_{k2}\right) \\ \hat{e}_{k2} = \frac{1}{\sqrt{2}}\left(+i\,\hat{\varepsilon}_{k1} + \hat{\varepsilon}_{k2}\right) \end{cases} \quad \left(-(i)^* = i\right)$$

we obtain two orthogonal <u>circular</u> polarizations.

$$\hat{e}_{k1} \times \hat{e}^*_{k2} = \frac{1}{2}\left(\hat{\varepsilon}_1 + i\hat{\varepsilon}_2\right) \times \left(-i\hat{\varepsilon}_1 + \hat{\varepsilon}_2\right) = \frac{1}{2}\left(\hat{\varepsilon}_1 \times \hat{\varepsilon}_2 + \hat{\varepsilon}_2 \times \hat{\varepsilon}_1\right) = 0$$

but
$$\hat{e}_{k1} \times \hat{e}^*_{k1} = \frac{1}{2}\left(\hat{\varepsilon}_1 + i\hat{\varepsilon}_2\right) \times \left(\hat{\varepsilon}_1 - i\hat{\varepsilon}_2\right) =$$
$$= \frac{1}{2}\left(-i\,\hat{\varepsilon}_1 \times \hat{\varepsilon}_2 + i\,\hat{\varepsilon}_2 \times \hat{\varepsilon}_1\right) = -i\,\hat{k}$$

$$\hat{e}_{k2} \times \hat{e}^*_{k2} = \frac{1}{2}\left(i\,\hat{\varepsilon}_1 + \hat{\varepsilon}_2\right) \times \left(-i\,\hat{\varepsilon}_1 + \hat{\varepsilon}_2\right) =$$
$$= \frac{1}{2}\left(i\,\hat{\varepsilon}_1 \times \hat{\varepsilon}_2 - i\,\hat{\varepsilon}_2 \times \hat{\varepsilon}_1\right) = i\,\hat{k}$$

(c) For general $d = |d|e^{i\varphi}$, we obtain <u>elliptic</u> polarization.

Useful identity

$$\sum_{\sigma=1,2} \left(\hat{e}^*_{k\sigma}\right)_i \left(\hat{e}_{k\sigma}\right)_j = \frac{1}{1+|d|^2}\Big\{\left(\hat{\varepsilon}_{k1} + d^* \hat{\varepsilon}_{k2}\right)_i \left(\hat{\varepsilon}_{k1} + d\, \hat{\varepsilon}_{k2}\right)_j$$

$$+ \left(-d\, \hat{\varepsilon}_{k1} + \hat{\varepsilon}_{k2}\right)_i \left(-d^* \hat{\varepsilon}_{k1} + \hat{\varepsilon}_{k2}\right)_j \Big\} =$$

$$= \frac{1}{1+|d|^2}\Big\{\left(\hat{\varepsilon}_{k1}\right)_i \left(\hat{\varepsilon}_{k1}\right)_j \left(1+|d|^2\right) + \left(\hat{\varepsilon}_{k2}\right)_i \left(\hat{\varepsilon}_{k2}\right)_j \left(|d|^2+1\right)$$

$$+ \left(\hat{\varepsilon}_{k1}\right)_i \left(\hat{\varepsilon}_{k2}\right)_j \left(d-d\right) + \left(\hat{\varepsilon}_{k2}\right)_i \left(\hat{\varepsilon}_{k1}\right)_j \left(d^*-d^*\right)\Big\}$$

$$= \left(\hat{\varepsilon}_{k1}\right)_i \left(\hat{\varepsilon}_{k1}\right)_j + \left(\hat{\varepsilon}_{k2}\right)_i \left(\hat{\varepsilon}_{k2}\right)_j = \sum_{\sigma=1,2} \left(\hat{\varepsilon}_{k\sigma}\right)_i \left(\hat{\varepsilon}_{k\sigma}\right)_j$$

But $\left(\hat{\varepsilon}_{k1}, \hat{\varepsilon}_{k2}, \hat{k}\right)$ form right-hand, orthogonal, Cartesian basis $\Rightarrow \left(\hat{\varepsilon}_{k1}\right)_i \left(\hat{\varepsilon}_{k1}\right)_j + \left(\hat{\varepsilon}_{k2}\right)_i \left(\hat{\varepsilon}_{k2}\right)_j + \hat{k}_i \hat{k}_j = \delta_{ij}$

(by the theorem of directional cosines)

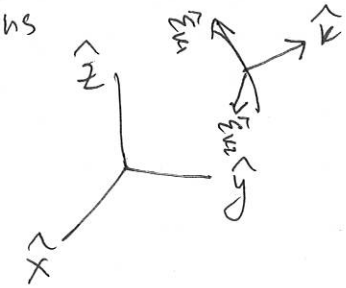
Therefore we proven that

$$\sum_{\sigma=1,2} \left(\hat{e}^*_{k\sigma}\right)_i \left(\hat{e}_{k\sigma}\right)_j = \delta_{ij} - \hat{k}_i \hat{k}_j \qquad (2)$$

For example, for $\hat{k} = \sin\theta \cos\varphi\, \hat{x} + \sin\theta \sin\varphi\, \hat{y} + \cos\theta\, \hat{z}$,

we have $\begin{cases} \hat{\varepsilon}_{k1} = \cos\theta \cos\varphi\, \hat{x} + \cos\theta \sin\varphi\, \hat{y} - \sin\theta\, \hat{z} \\ \hat{\varepsilon}_{k2} = -\sin\varphi\, \hat{x} + \cos\varphi\, \hat{y} \end{cases}$

two orthogonal <u>linear</u> polarizations



Tensor identity (2) can be checked explicitly now.

$$\varepsilon_{k1} = \cos\theta\cos\varphi\ \hat{x} + \cos\theta\sin\varphi\ \hat{y} - \sin\theta\ \hat{z}$$
$$\varepsilon_{k2} = -\sin\varphi\ \hat{x} + \cos\varphi\ \hat{y}$$
$$\hat{k} = \sin\theta\cos\varphi\ \hat{x} + \sin\theta\sin\varphi\ \hat{y} + \cos\theta\ \hat{z}$$

$$(\varepsilon_{k1})_i (\varepsilon_{k1})_j + (\varepsilon_{k2})_i (\varepsilon_{k2})_j = \delta_{ij} - \hat{k}_i \hat{k}_j$$

Check few cases:

let $i = j = 1$

$$\cos^2\theta\cos^2\varphi + \sin^2\varphi = 1 - \sin^2\theta\cos^2\varphi$$
$$\underbrace{(1-\sin^2\theta)}_{1-\sin^2\theta\cos\varphi}$$

Let $i=1$ $j=2$

$$\cos^2\theta\cos\varphi\sin\varphi - \sin\varphi\cos\varphi = -\sin^2\theta\cos\varphi\sin\varphi$$
$$\cos\varphi\sin\varphi\underbrace{(\cos^2\theta - 1)}_{-\sin^2\theta}$$

ok.

So $(\hat{\varepsilon}_{k1})_i (\hat{\varepsilon}_{k1})_j + (\hat{\varepsilon}_{k2})_i (\hat{\varepsilon}_{k2})_j + \hat{k}_i \hat{k}_j = \delta_{ij}$ is a statement of basis $\{\hat{\varepsilon}_1, \hat{\varepsilon}_2, \hat{k}\}$ completeness:

Multiply by a_i ($\vec{a} = (a_1, a_2, a_3)$) and b_j ($\vec{b} = (b_1, b_2, b_3)$) and sum over i, j:

$$\sum_i \left(a_i (\hat{\varepsilon}_{k1})_i\right)\sum_j\left(b_j (\hat{\varepsilon}_{k1})_j\right) + \sum_i\left(a_i (\hat{\varepsilon}_{k2})_i\right)\sum_j\left(b_j (\hat{\varepsilon}_{k2})_j\right) + \sum_{i,j}(a_i \hat{k}_i)(b_j \hat{k}_j)$$

$$= \sum_{i,j} a_i b_j \delta_{ij} = \sum_i a_i b_i = \vec{a}\cdot\vec{b}$$

$$\Rightarrow \vec{a}\cdot\vec{b} = (\vec{a}\cdot\hat{\varepsilon}_{k1})(\vec{b}\cdot\hat{\varepsilon}_{k1}) + (\vec{a}\cdot\hat{\varepsilon}_{k2})(\vec{b}\cdot\hat{\varepsilon}_{k2}) + (\vec{a}\cdot\hat{k})(\vec{b}\cdot\hat{k})$$